

Reward Repair and Connectivity Constraints for Cellular UAVs: Two Studies on a Barcelona Radio Map

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Research draft, 12 September 2026

Abstract—A cellular UAV must reach its destination while satisfying communication requirements. We investigate a master’s thesis reward formulation through two separately frozen reimplementations studies on a received Barcelona ray tracing map. Study I crosses reward replacement and arrival termination across four PPO treatments and five training seeds. Standard arrival rises from 21.4% for the legacy reward to 100.0% for reward replacement alone; termination alone achieves 0.0%. The combined change achieves 99.7% standard arrival but only 36.1% on longer routes. A subsequent audit shows why this is insufficient: only 92.2% of the reward only standard flights avoid every sampled RSS outage. Study II enforces minimum RSS and bounded queue occupancy, restores delay, interference and handover costs, and adds explicit resource choices and a shared prospective safety filter. Five seeds per arm yield 96.6% standard and 89.7% longer joint success with full radio cost, versus 96.9% and 86.3% without it. On paired successful standard flights, weighted radio cost falls 24.5%, but delay rises 186.7%. A deterministic radio selection reference achieves 98.0% and 92.5% joint success. The evidence supports a reward and endpoint repair within the declared surrogate, with retained negative findings and explicit thesis page comparisons; it establishes neither a PPO advantage nor reproduction of the original simulator.

Index Terms—Cellular UAV, reward shaping, connectivity constraints, handover, resource allocation, reinforcement learning, reproducibility.

I. INTRODUCTION

A 2026 master’s thesis uses a Barcelona ray tracing environment for joint UAV trajectory and handover control [1, pp. 6–7, Sec. 3.1]. Its objective combines delay, interference and handovers [1, p. 10, Eq. (11)], with minimum RSS and bounded queue constraints [1, p. 11, Eq. (12), C2 and C4]. The reward gives a fixed bonus whenever distance decreases [1, p. 12, Eqs. (13)–(14); p. 15, Table 3], while training continues after arrival to encourage braking [1, p. 12, Sec. 5.2]. The thesis reports wandering and prioritization of connectivity over destination completion [1, p. 19, Sec. 7.2; p. 21, Sec. 8]. Its listed observation omits velocity despite acceleration control [1, p. 12, Sec. 5.2; p. 9, Eq. (2)]. These are reasons to audit the formulation; they do not identify the exact behavior of unavailable source code. All thesis locators here use printed pages; add two to obtain the PDF viewer page.

Cellular control couples flight, association and resources. Related work jointly studies cargo UAV paths and association [2]; hybrid control [3], invalid action masking [4], and potential based shaping [5] provide established tools. We

contribute a controlled empirical diagnosis with reproducible artifacts, rather than a new algorithm or shaping theorem.

The two studies answer different questions. Study I isolates replacement of the written reward and termination on safe arrival within one shared surrogate. Its positive arrival result is then audited against the full communication task. Study II compares an arrival objective with the same objective plus all three communication costs under strict sampled RSS and queue constraints. Its controller and training range also change, so comparisons between studies are descriptive; only within study contrasts isolate their declared factors.

Both studies retain five training seeds per learned treatment, fresh test routes, deterministic references, unsuccessful designs and all reported failure outcomes. The original simulator and trained policies remain unavailable. We therefore make no claim of improvement over the thesis’s actual controller. The received private map has matching metadata, not verified experiment version identity. Detailed appendices distinguish original definitions, shared reimplementations choices, treatment factors, and the subsequent connectivity repair. Favorable aggregate communication cost never replaces the primary mission outcome or the individual radio components.

II. DATA AND INITIAL SHARED SURROGATE

The following model defines Study I and the common physical assumptions retained in Study II. Its permissive connectivity endpoint is superseded in Study II, as described in Section V.

A. Received Radio Map

The file `Barcelona_dataset_January.h5` contains an $133 \times 5000 \times 3500$ signed 8 bit RSS array, station identities and coordinates, operator membership, and transmit powers. Its metadata gives 2.1 GHz, 100 m altitude, and nominal 1 m grid resolution. The file size is 2,327,593,160 bytes. We use the 87 station operator corresponding to the thesis’s first operator [1, p. 14, Sec. 6.1]. The station counts, frequency, height, and dimensions match the description [1, p. 6, Sec. 3.1.1; p. 7, Table 2], but exact experiment version identity is not established.

The original stored coordinate vectors span 0–5000 m with 5000 samples and 0–3500 m with 3500 samples. Their spacings are approximately 1.00020004 m and 1.00028580

m. Nearest neighbor lookup uses these vectors; all training and evaluation use native resolution. Values are promoted to floating point before radio arithmetic. The untouched source checksum and schema are retained.

A full scan finds that 27.77245% of all link entries equal -128 , while the other entries range from -110 to -25 dBm. We provisionally interpret -128 as an absent usable path and assign it zero received power. The generator's sentinel and quantization conventions still require confirmation. For the selected operator, every grid position has at least 38 stations above -96 dBm, and the strongest RSS ranges from -51 to -25 dBm. Thus the stored map has no inevitable geographic coverage hole under this RSS threshold. This observation does not guarantee a permissible handover or adequate SINR.

B. Motion, Mission, and Observations

The thesis writes scalar speed and a selected direction in its position update [1, p. 9, Sec. 3.2, Eq. (2)]. Our shared benchmark instead preserves an inertial velocity vector at constant altitude. With velocity projection,

$$\mathbf{v}_{t+1} = \Pi_{\|\mathbf{v}\| \leq v_{\max}}(\mathbf{v}_t + \mathbf{a}_t \Delta t), \quad (1)$$

$$\mathbf{p}_{t+1} = \mathbf{p}_t + \frac{1}{2}(\mathbf{v}_t + \mathbf{v}_{t+1})\Delta t. \quad (2)$$

The policy supplies radial and lateral acceleration components in a frame aligned with the current destination bearing. Their norm is limited to 5 m/s^2 , and speed is limited to 25 m/s . This continuous action interface replaces the five acceleration bins and eight headings [1, p. 12, Sec. 5.2; p. 15, Table 3]. The thesis uses a 100 m/s penalty threshold [1, p. 15, Table 3]; ours is a hard 25 m/s cap. The step is 1 s and the deadline is 200 s . The source gives 1 ms in prose [1, p. 14, Sec. 6.1] and 0.1 s in Table 3 [1, p. 15]; we explicitly choose a new time discretization.

Safe arrival is the common event

$$\mathcal{G}_t = \{\|\mathbf{p}_t - \mathbf{g}\| \leq 10 \text{ m}, \|\mathbf{v}_t\| \leq 2 \text{ m/s}\}. \quad (3)$$

The 10 m position tolerance is retained from Table 3 [1, p. 15]; the numerical 2 m/s terminal speed test is our addition to the stated stopping intention [1, p. 9, Sec. 3.2]. A boundary crossing, battery exhaustion, or five consecutive RSS outage samples ends a failure. The thesis explicitly terminates on energy exhaustion and penalizes boundary, RSS, and speed violations [1, p. 12, Sec. 5.2, Eq. (14)]; our boundary and repeated outage terminations are additional rules. Speed is constrained directly. Building geometry supports propagation in the thesis [1, p. 6, Sec. 3.1.1]. Our received RSS array does not provide a collision model, and obstacle safety is outside the evaluated model; we do not assume the original code enforced it.

Every treatment receives the same 42 dimensional observation: position; destination bearing and distance; radial, lateral, and total velocity; remaining time encoded by elapsed time; energy; buffer; prior arrival flag; resource and load phases; network decision phase; serving SINR; outage count; and candidate RSS, relative positions, identities, and masks. This

replaces the listed six component vector of position, serving station, resource group, energy, and buffer [1, p. 12, Sec. 5.2]. Observation repair is shared, and its independent effect is not estimated.

C. Association and Communication Proxies

A categorical policy selects stay or one of the four strongest candidate stations. A request is admissible when candidate RSS exceeds both -96 dBm and serving RSS plus 3 dB . Network actions occur every five seconds, with an additional opportunity on an observed serving RSS outage. This emergency rule was introduced during validation, before training, when a periodic decision could otherwise arrive after the outage failure deadline. We do not implement or claim a complete 3GPP time to trigger procedure. Masking follows the standard practice of excluding invalid categorical actions [4]. The inequality follows the thesis's A3 equation [1, p. 9, Eq. (3)], with the numerical offset from Table 3 [1, p. 15]; the five action candidates and decision interval are our choices. The source greedy controller uses viability counters [1, pp. 13–14, Sec. 5.4], which we do not copy.

There are twelve resource groups and six occupied background groups per station in a fixed spatial pattern. A shared first available allocation rule chooses the resource group; resource allocation is not learned. Scenario phase rotates group labels globally and does not supply independent traffic dynamics. The source permits learned RBG requests [1, p. 9, Sec. 3.3.1; p. 12, Sec. 5.2] and describes random occupancy of at least half the groups [1, p. 14, Sec. 6.1]; both differ from our shared allocator. Since the source RSS is a downlink budget [1, p. 7, Eq. (1)], station powers are aggregated as a downlink interference proxy:

$$P_k = 10^{(\text{RSS}_k - 30)/10}, \quad \text{SINR} = \frac{P_s}{N + \sum_{k \neq s} o_k P_k}. \quad (4)$$

Here powers are in watts, o_k indicates background occupancy of the selected group, and noise including the noise figure is -103.41 dBm . The ideal service rate is $1.44 \cdot 10^6 \log_2(1 + \text{SINR}) \text{ bit/s}$. It is neither measured throughput nor a validated uplink interference model. This replaces the printed bandwidth-plus-logarithm rate, SNR in dB, and sum of RSS values labeled uplink interference [1, p. 10, Eqs. (7)–(9)]. It changes the interference model as well as correcting power arithmetic; we cannot establish whether the original code used the printed expressions.

Deterministic offered traffic is 200 kbit/s , buffer capacity is 1.28 Mbit , and a handover adds 4800 bits . The control overhead and nominal buffer capacity follow Table 3 [1, p. 15], interpreting KB as decimal kilobytes. Deterministic traffic replaces the Poisson formulation [1, pp. 9–10, Eqs. (5)–(6)]; the thesis also gives different data packet sizes in prose and table [1, pp. 14–15, Sec. 6.1, Table 3]. The backlog/service delay retains the ratio in Eq. (4) [1, p. 9], but we explicitly use postservice backlog and censor at 200 s . Energy is the integral of $100 + 0.4\|\mathbf{v}\|^2$ watts against a 100 kJ budget. This explicit proxy replaces the mass, efficiency, lift-to-drag and electronics

TABLE I
SHARED REIMPLEMENTATION CHOICES, HELD FIXED ACROSS
TREATMENTS

Aspect	Declared benchmark choice
Observations	Goal and velocity observed; 42 dimensions
Motion	Continuous acceleration, 1 s steps, 25 m/s cap
Network	Stay plus four candidates; masks; outage requests
Resources	Common deterministic first available allocator
Interference	Linear downlink cochannel power proxy
Traffic and energy	Deterministic traffic; explicit energy proxy
Source equivalence	Original code and checkpoints unavailable

expression [1, p. 10, Eq. (10)] and the capacity labeled 1000 kW [1, p. 15, Table 3]. It does not claim a calibrated rotorcraft model. Table I summarizes the main departures from the source description.

III. STUDY I: REWARD AND ARRIVAL TERMINATION

A. Why the Binary Approach Bonus Is Suspect

Let d_t be destination distance, D_t the declared delay proxy, I_t the linear interference proxy, and h_t a handover indicator. The legacy reward treatment implements the structure in Eqs. (13)–(14) [1, p. 12], with the equal-policy weights, scales, and bonus in Table 3 [1, p. 15]:

$$r_t^L = \frac{.35}{1 + 10D_t} + \frac{.30}{1 + 10^5 I_t} + \frac{.35}{1 + 100h_t} + 12\mathbb{I}[d_{t+1} < d_t] - 10O_t - 10B_t, \quad (5)$$

where O_t and B_t mark RSS outage and a boundary crossing. Energy exhaustion overrides the reward with -10 . The shared speed constraint replaces a soft speed penalty. The interference input is corrected to watts for all treatments; thus this is an ablation of the written incentive, not literal replication of potentially erroneous source equations.

The approach term depends on the sign of progress but not its magnitude. A controller can receive the same bonus for very slow progress as for efficient flight. A retreat carries no matching distance cost, permitting positive approach bonuses on a closed excursion outside the destination region. Moreover, stopping the episode without a compensating mission objective can remove future positive reward and reinforce avoidance of completion. Similar failures motivate potential based shaping [5].

B. Mission Reward and Terminal Potential

Define the potential, with $L = 100$ m,

$$\Phi(s) = -\frac{d + \|\mathbf{v}\|^2 / (2a_{\max})}{L}. \quad (6)$$

The speed term is a stopping distance scale. The replacement reward is

$$r_t^M = 20A_t - 20F_t - 0.05 - 0.05c_t + \gamma\Phi(s_{t+1}) - \Phi(s_t), \quad (7)$$

where A_t marks first safe arrival, F_t a terminal failure before any safe arrival, and $c_t = \min(D_t, 1) + O_t + h_t$. The prior

TABLE II
FACTORIAL TRAINING TREATMENTS

Label	Reward	Terminate on arrival
Legacy	r^L	No
Termination only	r^L	Yes
Reward only	r^M	No
Both	r^M	Yes

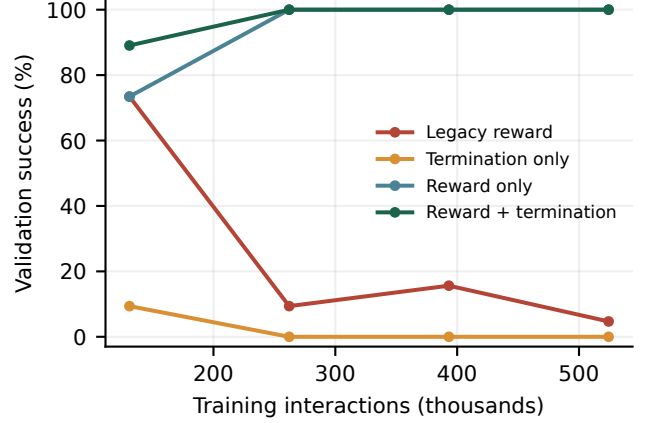


Fig. 1. Pilot seed 42 on 64 validation routes. Curves are used to document development and budget choice, not to substitute for the frozen test comparison.

arrival flag is observed, and the success bonus is paid only once even if training continues afterward. Set the potential to zero at every terminal, including the deadline. For any episode ending at T , the shaping contribution satisfies

$$\sum_{t=0}^{T-1} \gamma^t [\gamma\Phi(s_{t+1}) - \Phi(s_t)] = -\Phi(s_0). \quad (8)$$

Consequently, within each fixed terminal formulation, shaping preserves the discounted base objective [5]. This identity is checked numerically. It does not imply that the finite weighted base reward certifies every physical constraint or guarantees successful learning. The identity concerns raw rewards. Running normalization and clipping in PPO can change the effective update objective; this experiment tests the declared learning recipe without isolating those interactions.

C. Common PPO Implementation

All arms use proximal policy optimization [6] with separate actor and critic networks, each containing two 64 unit tanh layers. The actor has a Gaussian motion head and a masked categorical association head. Motion samples are transformed through tanh and the acceleration norm limit; likelihood ratios are consistently computed in latent action space. The entropy regularizer sums latent Gaussian and masked categorical entropy.

Training uses 64 parallel lanes, 128 step rollouts, four epochs per rollout, 512 sample minibatches, Adam learning rate $3 \cdot 10^{-4}$, clip 0.2, $\gamma = 0.99$, GAE parameter 0.95, entropy

coefficient 0.005, value coefficient 0.5, and gradient norm limit 0.5. Rewards are divided by a running standard deviation of discounted returns without subtracting a mean, and clipped to $[-10, 10]$, consistently across arms. Finite horizon terminals are not bootstrapped. No pretrained model, demonstrations, or imitation loss is used. The thesis uses Stable Baselines [1, p. 13, Sec. 5.3]; our PPO is a new native implementation. Table VIII contrasts every reported source setting with ours [1, p. 16, Table 4]. For example, learning rate increases from $3 \cdot 10^{-5}$ to $3 \cdot 10^{-4}$, entropy weight decreases from 0.01 to 0.005, and the source’s 0.03 target KL is replaced by diagnostic KL logging without an early stopping threshold.

D. Validation, Freeze, and Held Out Routes

Training uses 4096 routes sampled with lengths 200–1000 m, with 64 separate validation routes. The final standard set contains 200 new routes of the same length range. A second set contains 200 routes of length 1000–1800 m. These are unseen coordinate pairs in the same city, not independent cities or held out geographic regions. Starts and destinations stay inside documented map margins. All coordinates, seeds, and background phases are saved. The original tuning and comparison sections do not specify three independent training, validation, and test route sets [1, pp. 15–16, Secs. 6.2–6.3]. This is a limit of the written protocol, not proof that its unavailable code lacked such a separation.

Six pilots, using seeds 41 and 42, established learning behavior and the 524,288 interaction budget. All final arms use seeds 1101–1105 and exactly that budget, totaling 10,485,760 interactions. Source hashes and configuration were frozen before trained policy test evaluation. The initial feasibility inspection used separate exploratory route sets, which were retired before training; final route seeds are 42012 and 42013. Final checkpoints are evaluated without early stopping or test based selection. Each lane follows the same ordered training scenario stream across treatments, although successful termination changes how many episodes fit in an equal interaction budget.

The evaluated continuous action is the Gaussian mean after transformation, and the network action is the masked categorical mode. Every policy’s evaluation stops at first safe arrival, independently of its training termination rule. Thus the continuing training treatment is still evaluated as a finite mission.

E. Metrics, Reference, and Uncertainty

Study I arrival success is primary. Its historical recorded feasible success additionally requires no packet overflow and an RSS outage fraction at most 5%. This limited feasibility label does not imply obstacle or airframe certification. Failure counts and final distances are retained. Time, path, energy proxy, delay, interference, SINR, and handovers are reported conditional on success with their sample counts, avoiding aggregation of failed flights into radio gains.

The deterministic reference points its desired velocity toward the goal, uses a braking schedule based on stopping

distance, and requests the strongest admissible station. It shares every environmental constraint and resource rule. It establishes that the tested missions are attainable and provides a practical nonlearning comparison; it is not asserted to be globally optimal.

We show all five seed outcomes and standard deviations. The primary contrast is both changes minus legacy success on standard routes. A 10,000 draw crossed bootstrap resamples the five seed identities and 200 route identities, using the same sampled identities for both arms. This preserves pairing and avoids counting 1000 episodes as 1000 independent training runs. Other factorial and longer route contrasts are secondary. Interval reporting follows the broader need to expose uncertainty in few run RL evaluations [7]. Five seeds still provide limited resolution and uncertain tail behavior.

The prespecified practical gate for a positive repair finding requires at least 90% mean standard mission success, at least 90% recorded feasible success, and a positive lower 95% bootstrap bound for the primary contrast. It is a reporting gate rather than a claim of statistical or deployment certification.

F. Completion and Factorial Effects

Table III and Fig. 2 present the full frozen comparison. Reward replacement alone achieves 100.0% standard success compared with 21.4% for the legacy reward. The combined change achieves 99.7%, while termination alone achieves 0.0%. The primary gain is 78.3 percentage points, with a paired bootstrap interval [66.2, 89.5]. Recorded feasible success for the combined change is 97.8%, satisfying the practical arrival gate. This gate did not require uninterrupted sampled connectivity. These results identify the replacement reward as the essential tested intervention; changing arrival termination alone does not resolve the failure.

The pilots also illustrate why longer training cannot automatically repair a misaligned objective. In pilot seed 42, legacy validation success initially rises to 73.4% and then falls to 4.7% by the final budget. Reward replacement reaches 100% by approximately 262,144 interactions and retains it through the pilot’s final checkpoint. Figure 1 is explicitly exploratory validation evidence from one pilot, not a confidence interval over final seeds.

G. Longer Routes and the Role of Termination

Reward replacement with continuing training reaches 100.0% success on longer routes. The combined treatment reaches 36.1% with seed standard deviation 12.6 percentage points. Thus the treatment that provides efficient completion within the training range does not necessarily generalize best to longer flights. Training termination changes episode exposure and visitation distribution even under equal interaction budgets, but the experiment does not isolate the mechanism causing this extrapolation difference. The longer route weakness is retained rather than removed by further test driven tuning.

Figure 3 shows the first predefined standard test route for seed 1101, selected by index rather than performance. Its

TABLE III

FROZEN RESULTS: PERCENTAGES ARE MEAN $\pm$ STANDARD DEVIATION OVER FIVE TRAINING SEEDS. STANDARD ROUTES ARE 200–1000 M; LONGER ROUTES ARE 1000–1800 M. EACH LEARNED POLICY EVALUATES 200 ROUTES PER SPLIT.

Treatment	Mission success (%)		Recorded feasible success (%)		Successful episodes	
	Standard	Longer	Standard	Longer	Standard	Longer
Legacy	21.4 $\pm$ 14.3	0.0 $\pm$ 0.0	20.8 $\pm$ 13.9	0.0 $\pm$ 0.0	214/1000	0/1000
Termination only	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0/1000	0/1000
Reward only	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	98.7 $\pm$ 1.4	96.0 $\pm$ 1.1	1000/1000	1000/1000
Both	99.7 $\pm$ 0.7	36.1 $\pm$ 12.6	97.8 $\pm$ 0.8	35.2 $\pm$ 12.6	997/1000	361/1000
Deterministic reference	100.0	100.0	97.0	95.5	200/200	200/200

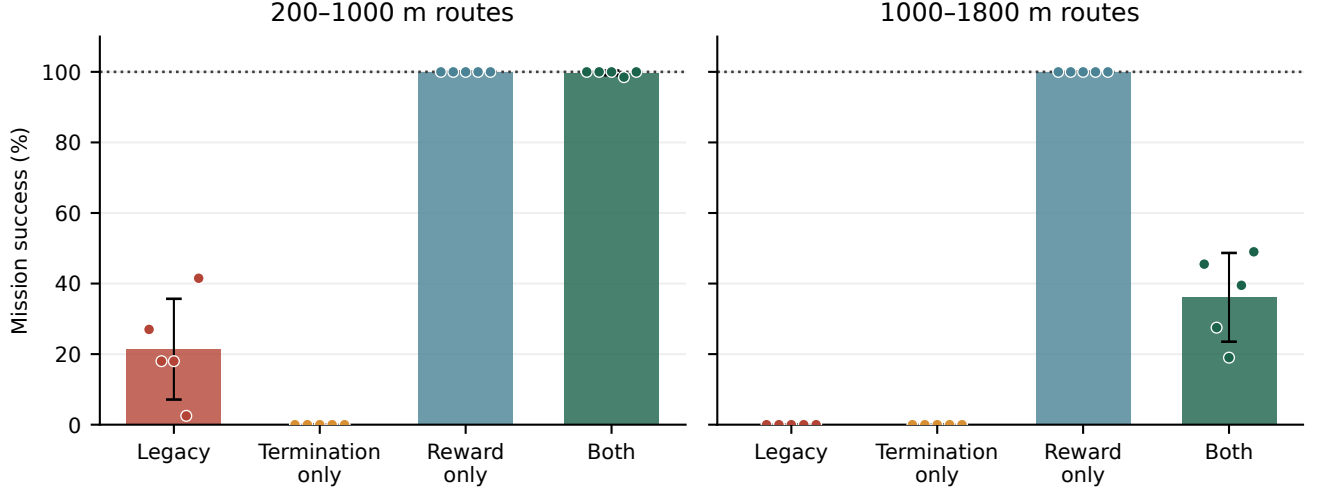


Fig. 2. Study I arrival success under its permissive connectivity rules. Each dot is one training seed, bars show the mean, and error bars show seed standard deviation. There are 200 shared routes per split and five seeds per learned treatment. The dotted line is the deterministic reference. No test outcome was used to tune the frozen policies.

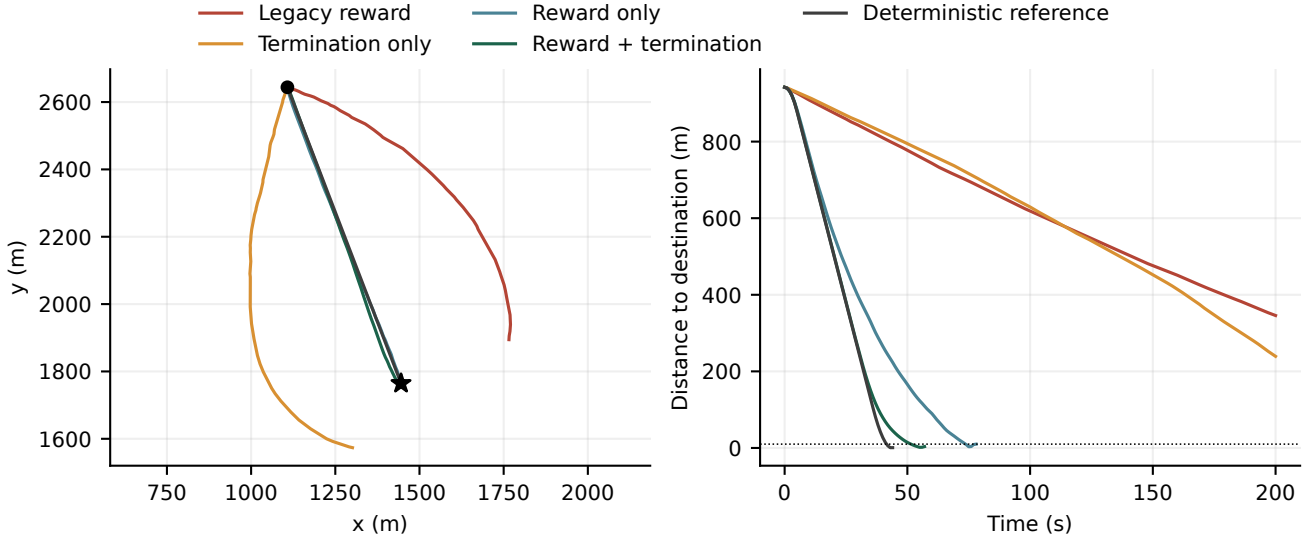


Fig. 3. First standard test route and first declared training seed. The black circle and star denote start and destination. All learned treatments and the deterministic reference share the scenario. The dotted distance line marks 10 m; low speed is also required for success.

trajectories and distance traces illustrate the different learned behavior, while the aggregate tables remain the basis for the

conclusions. Proximity by itself is insufficient: the evaluated success criterion also requires low speed.

TABLE IV

STANDARD ROUTE METRICS CONDITIONAL ON MISSION SUCCESS, POOLED ACROSS THE FIVE SEEDS. THE DETERMINISTIC REFERENCE HAS ONE RUN ON 200 ROUTES. DIFFERENT SUCCESSFUL SUBSETS PREVENT UNQUALIFIED SECONDARY DOMINANCE CLAIMS. ENERGY, DELAY, AND INTERFERENCE ARE THE DECLARED PROXIES.

Treatment	n	Time (s)	Energy (kJ)	Handovers	SINR (dB)	Delay (s)	Outage (s)	Interf. (μ W)
Legacy	214	141.25	14.80	8.83	-6.34	0.88	0.07	0.70
Termination only	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Reward only	1000	50.23	8.29	3.96	-7.91	2.58	0.09	0.74
Both	997	36.52	8.17	2.99	-8.13	2.83	0.09	0.74
Deterministic reference	200	28.62	7.99	2.38	-8.51	3.45	0.09	0.74

H. Efficiency and the Deterministic Reference

The deterministic reference achieves 100.0% standard and 100.0% longer route success. Among successful standard missions, mean flight times are 141.2 s for legacy, 50.2 s for reward only, 36.5 s for both changes, and 28.6 s for the reference. Because the successful subsets differ, these means alone are not paired efficiency comparisons. On matched successful standard routes, the combined policy takes 7.9 s more time than the reference on average. The evidence establishes repair of a learned objective, not an advantage over the deterministic controller.

Table IV reports success conditional secondary metrics with sample counts. The radio summaries are downlink proxies and the energy measure is uncalibrated. A controller with lower interference or delay on a subset of completed short routes is not ranked above a controller that completes the whole route set. Raw per episode records permit matched subset analysis without concealing failures or mixing physical units. The original results emphasize SNR, outage, interference, remaining energy, and handover count [1, pp. 16–20, Sec. 7, Figs. 5–8]. Our success rate, terminal speed and distance checks, success conditional reporting, and paired seed uncertainty are additional evaluation choices.

IV. CONNECTIVITY REQUIREMENTS AND THE STUDY I AUDIT

A. Which Communication Requirements Apply?

We use the thesis requirements, not a newly imposed application service level. The RSS minimum is -96 dBm and the buffer is listed as 160 KB [1, p. 15, Table 3]; we retain the explicit 1,280,000 bit interpretation. The written C2 inequality allows equality and applies at every modeled time [1, p. 11, Eq. (12)]. Delay remains an optimized quantity, defined in the source as queue divided by service rate [1, p. 9, Eq. (4)]. Neither a two second packet deadline nor an empty queue arrival condition is introduced. A successful arrival can therefore retain queued data; it must not be described as completed data delivery.

The distinction between formulation and enforcement matters. The source reward penalizes RSS violations [1, p. 12, Eq. (14)], which does not establish that its unavailable implementation guaranteed C2. Its direct data dump penalty was removed during development [1, pp. 20–21, Sec. 7.3]. We

compare verified written definitions with our executable rules, not inferred behavior of missing original code.

B. Why the Earlier Arrival Result Was Insufficient

The original map audit found at least 38 operator 1 stations strictly above -96 dBm at every grid location. The strongest station ranges from -51 to -25 dBm. Consequently, an RSS coverage hole does not force a detour if any suitable station could be selected. Actual association constraints and interference still matter.

An exact replay of 4,400 Study I flights added service diagnostics without changing their saved outcomes. In the earlier reward replacement's 1,000 standard evaluations, arrival was 100%, but only 92.2% had no sampled RSS outage. Modeled capacity was below the 200 kbit/s offered rate for a mean 26.6% of each flight, and 20.2% arrived with a nonempty queue. A transient capacity deficit is not itself disconnection because buffering can absorb it. These measures expose what the old endpoint did and did not establish; they are not a new confirmatory policy comparison.

V. STUDY II: STRICT SAMPLED CONNECTIVITY

Study II retains the map, motion, radio arithmetic, traffic and energy assumptions in Section II: 1 s steps, 5 m/s² acceleration, 25 m/s maximum speed, 200 s deadline, 100 kJ energy budget, 12 resource groups, static half occupancy, and 200 kbit/s offered traffic. These remain explicit surrogate choices. The changes below concern the mission endpoint, radio objective and control architecture, not a recovered original simulator.

A. Joint Success and Failure

Let d_t denote destination distance and q_t the queue after the sampled arrival/service update. Joint success requires arrival by the deadline with $d_T \leq 10$ m and $\|v_T\| \leq 2$ m/s, together with

$$\text{RSS}_{k_t}(p_t) \geq -96 \text{ dBm}, \quad 0 \leq q_t \leq 1,280,000 \text{ bits}, \quad \forall t. \quad (9)$$

The 10 m tolerance follows the source table [1, p. 15, Table 3]; the numeric stopping tolerance is our explicit interpretation of stopping at the destination [1, p. 9, Sec. 3.2]. Map and energy conditions also apply. A single sampled RSS violation or any buffer overflow ends a failed mission. Failure takes precedence over simultaneous arrival. Initial RSS is checked; queue clipping for storage cannot erase an overflow flag.

This enforces the constraints at modeled samples. It is not a guarantee between the one second samples or within the queue update. No minimum SINR, packet latency threshold, or final queue draining requirement is claimed.

B. Actions, Residual Motion and Safety Filter

The source includes BS and RBG actions [1, p. 12, Sec. 5.2]. We expose 61 categorical options: stay, or five candidate station slots by 12 RBGs. Candidates are serving plus the four strongest RSS stations. Masks reject occupied groups and duplicated serving station slots. A new station must have current RSS more than 3 dB above the serving RSS and meet the minimum, following the source's A3 direction [1, p. 9, Eq. (3); p. 15, Table 3]. Network decisions occur each modeled second, replacing the earlier surrogate's five second interval. Resource changes within the same station are allowed and counted separately from handovers. Handovers generate 4,800 control bits [1, p. 15, Table 3]; switching interruption is not modeled.

The observation has 156 features covering geometry, velocity, time, queue, energy, serving SINR, candidate RSS and positions, option capacities and masks. Two Gaussian outputs define bounded residual acceleration commands. We project a destination directed braking command onto the unit disk, add half of the tanh transformed residual, and apply the shared acceleration projection. Both learned arms use this prior; they do not learn basic flight entirely from scratch. The source's fully discrete action and smaller state differ [1, p. 12, Sec. 5.2].

A shared safety filter predicts the proposed next position, RSS and queue. It first tries another admissible station/resource pair for the same motion. If none is feasible, it invokes the local motion/network search. If no candidate repairs the step, the flight fails normally. The filter has prospective access to the known map and is an explicit control component, not learned competence. Interventions are recorded. It cannot establish global reachability.

C. Restoring the Communication Objective

Using source equal weights and scales [1, p. 15, Table 3], define

$$C_t = 0.35 \frac{10D_t}{1 + 10D_t} + 0.30 \frac{10^5 I_t}{1 + 10^5 I_t} + 0.35 \frac{100H_t}{1 + 100H_t}. \quad (10)$$

Here D_t is the backlog/service delay proxy in seconds, I_t interference in watts, and H_t a handover indicator. These are one minus the source reciprocal utilities [1, p. 12, Eq. (13)], removing the positive reward for merely remaining active while retaining all communication components. The earlier replacement's flat delay penalty above one second is removed. The reciprocal cost still saturates smoothly, which can favor substantial delay in exchange for fewer handovers; it is not a latency guarantee.

Both learned arms use

$$\Phi_t = -\frac{d_t + \|v_t\|^2/(2a_{\max})}{100 \text{ m}}, \quad (11)$$

with terminal potential zero. The base reward is $0.99\Phi_{t+1} - \Phi_t - 0.05$, plus 20 for joint success and minus 20 for terminal failure. The arrival arm uses this base alone; the full arm subtracts C_t . Both have identical hard failure rules and control interfaces. This is not a comparison to the original trained thesis policy or a claim that a finite scalar reward proves optimal lexicographic control.

VI. STUDY II PROTOCOL AND RESULTS

A. Learning, Splits and Freeze

PPO retains all optimizer, network width, rollout, normalization and discount settings of Study I, with the categorical head expanded from five to 61 logits. Likelihoods refer to proposed residual and network actions; projection and filtering are environment transformations.

Training has 4,096 routes of lengths 200–1800 m. Validation has 64 standard and 32 longer routes. Fresh tests contain 200 standard routes of 200–1000 m and 200 longer routes of 1000–1800 m. Exact route pairs do not overlap with the earlier study or other v2 splits, but all use the same geographic map. Both length ranges occur during training, so the longer set is not an extrapolation test. The source does not document equivalent three way separation [1, pp. 15–16, Secs. 6.2–6.3].

Six development policies and unsuccessful local planning variants are retained in Appendix D; their outcomes use validation routes only. Final source, scenario and protocol hashes were frozen before test evaluation. Each arm uses seeds 2101–2105 and 524,288 interactions, totaling 5,242,880 final interactions across ten policies. Evaluation uses the final checkpoint and deterministic actions; no test result selects a checkpoint or controller revision.

B. References and Analysis

Four deterministic references share the same filter and rules: nominal straight flight with strongest RSS association (Goal RSS); the same nominal motion with prospective station/resource cost minimization (Goal radio); a one step joint search; and a three step joint search. Joint searches use nine initial motion primitives: direction offsets and speed reductions of the braking reference. They prioritize sampled feasibility and progress, then communication cost. The three step search follows each initial choice with two predicted reference steps. Both are local methods and may fail. The filter can alter nominally straight motion, which is reflected in actual paths and intervention counts.

Primary reporting includes every mission and its failure reason. Secondary communication comparisons use the intersection of successful paired routes; intersection sizes and accumulated and average costs are retained. Paired differences use 5,000 crossed bootstrap draws over seed identities and shared route identities; deterministic comparisons resample routes only. Five training seeds do not represent independent geographic or traffic realizations. The prespecified strong joint improvement claim requires a positive lower interval for

TABLE V
JOINT SUCCESS AND SUCCESSFUL STANDARD MISSION COST. LEARNED
ROWS AGGREGATE FIVE SEEDS ON 200 ROUTES; REFERENCES USE 200
ROUTES.

Controller	Standard (%)	Longer (%)	$\sum C_t$
PPO arrival	96.9	86.3	6.57
PPO full	96.6	89.7	4.97
Goal RSS	97.0	87.0	6.22
Goal radio	98.0	92.5	3.23
Joint one step	96.5	90.0	2.62
Joint three steps	95.0	90.5	1.92

completion improvement and a negative upper interval for accumulated communication cost change. Individual components are reported regardless.

C. Strict Joint Completion

Table V and Fig. 4 report the frozen results. The full learned arm achieves 96.6% standard and 89.7% longer joint success, compared with 96.9% and 86.3% for the arrival arm. The paired standard completion difference is -0.3 percentage points, with 95% interval [-2.6, 2.0]. The corresponding longer difference is 3.4 points [-1.0, 8.1].

Both completion intervals include zero. The prespecified strong joint improvement claim is therefore not met on either split, despite lower weighted communication cost. Goal radio has the highest observed completion among the tested controllers: 98.0% standard and 92.5% longer. These results do not establish a PPO advantage over the simpler reference.

Every reported success has zero observed RSS violations and zero overflow, as required by the new endpoint. This does not mean every evaluated flight succeeds. The full arm has 34 failed standard evaluations and 103 failed longer evaluations out of 1,000 in each set. The standard failures are 32 RSS violations and two overflows; the longer failures are 77 RSS violations and 26 overflows. Arrival only has 31 standard RSS failures and 136 longer RSS failures plus one overflow. All per seed outcomes and deterministic failure counts remain in the analysis artifacts.

D. Communication Tradeoffs

There are 945 paired successful standard evaluations and 802 longer evaluations for the learned comparison. On those pairs, accumulated communication cost changes by -24.5% and -14.7%, respectively. Table VI shows the individual components and uncertainty. An improvement in the weighted objective is not equivalent to improvement in every metric. In particular, the standard mean delay proxy changes by 186.7%, while handovers change by -95.5%.

Goal radio improves longer completion over Goal RSS by 5.5 points [1.5, 9.5] and reduces accumulated cost by 41.0% on 171 paired successful routes. Three step planning reduces cost by a further 42.9% relative to Goal radio on 173 longer pairs, but its completion difference of -2.0 points [-6.5, 2.5] does not establish improvement.

On the matched standard routes, the full arm’s delay proxy is 4.21 s versus 1.47 s for the arrival arm, while handovers fall from 8.55 to 0.38 per mission. On matched longer routes, delay is 6.62 s versus 1.83 s. The unweighted interference difference is inconclusive on both splits. These component results limit the interpretation of the favorable aggregate cost.

VII. VALIDITY, REPRODUCIBILITY AND IMPLICATIONS

The initial test suite contains 34 passing checks covering movement, speed and acceleration bounds, position and speed arrival conditions, one time success payment, terminal potential telescoping, finite horizon GAE, mask validity, likelihood consistency, radio axes and spacing, emergency handovers, identical dynamics across reward arms, deterministic completion, and seed repeatability. Reloading the first declared reward only checkpoint reproduces all 200 recorded standard test episode records exactly. Passing these checks validates important properties of this code, not equivalence to unavailable source code.

Experiments use Python 3.12, CPU PyTorch 2.4.1, NumPy 2.3.5, and h5py 3.16.0 on a Windows machine with 16 logical processors and approximately 32 GiB RAM. Study I recorded approximately 9.4 minutes of training and final validation; Study II recorded 865.68 s. Both exclude map loading and final test evaluation. The total final training budget is 15,728,640 interactions across 30 policies; pilot budgets are separate. Checkpoints, all configurations, route splits, training logs, per episode metrics, source hashes, analysis scripts, and figures are retained locally. The received map is unchanged and excluded from version control. Independent reproduction currently requires obtaining that private artifact from its owner; no public dataset release is claimed.

Several limitations constrain interpretation. First, the shared full state, continuous actions, and network masks mean the results are not a direct comparison with the thesis’s partially observed discrete controller. Second, the radio, scheduling, delay, and energy models include declared simplifications; the full original simulator and exact dataset version remain unverified. Third, training and testing share one city, one height, one frequency, and one fixed load pattern. Fourth, five training seeds cannot establish robust tail behavior. Fifth, the controller is evaluated with deterministic actions and a fixed training budget, so the measured comparison includes these choices. Finally, the deterministic reference is already highly effective; the study does not justify adding a learning controller for this setting on efficiency grounds alone.

Study II adds 17 behavioral checks for strict RSS and overflow handling, resource masks, radio costs, the safety filter, action projection, lookahead isolation, and repeatable PPO parameter updates. All 51 tests pass. Every one of the 5,600 Study II final evaluations replays exactly; the separate 4,400 flight Study I connectivity audit preserves its original outcomes. The six Study I and seven Study II frozen source/protocol hashes remain unchanged. These checks establish implementation consistency, not physical validation.

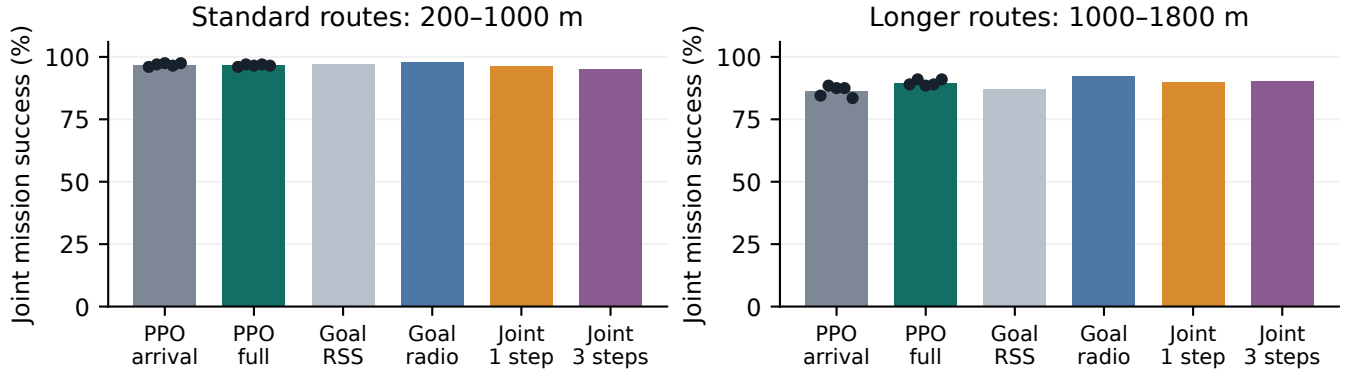


Fig. 4. Strict sampled joint success on fresh routes. Dots show individual training seeds for learned controllers. Deterministic controllers have one evaluation per route, not five independent replications.

TABLE VI
FULL MINUS ARRIVAL ON PAIRED SUCCESSFUL MISSIONS. DIFFERENCES AND 95% CROSSED BOOTSTRAP INTERVALS. NEGATIVE COST, TIME, DELAY, INTERFERENCE AND HANDOVER CHANGES FAVOR FULL.

Metric	Standard difference [95% interval]	Longer difference [95% interval]
Accumulated radio cost	-1.610 [-2.016, -1.230]	-2.393 [-3.317, -1.488]
Mean radio cost	-0.060 [-0.075, -0.046]	-0.039 [-0.054, -0.024]
Arrival time (s)	-0.195 [-0.267, -0.124]	-0.353 [-0.552, -0.168]
Delay proxy (s)	2.738 [2.030, 3.474]	4.789 [4.180, 5.397]
Interference (μ W)	0.007 [-0.005, 0.018]	-0.002 [-0.012, 0.008]
Handovers	-8.165 [-9.226, -7.155]	-19.562 [-21.719, -17.263]
Mean SINR (dB)	-1.095 [-1.374, -0.801]	-1.581 [-1.780, -1.393]

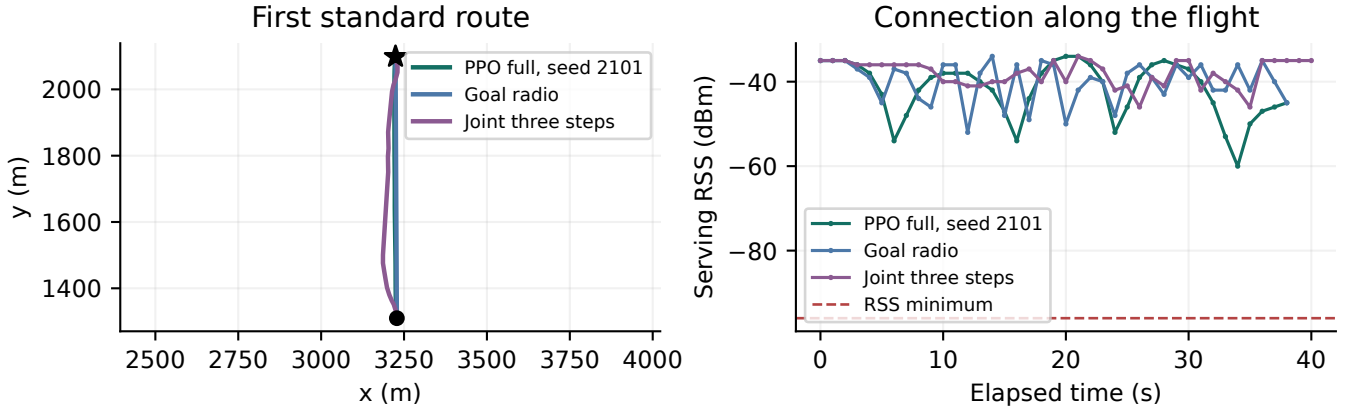


Fig. 5. The first declared standard test route and first full reward training seed, shown with two model based references. The displayed RSS includes the initial and terminal sample. This is a fixed example, not a selected typical route or proof of connectivity between samples.

The frozen Study II estimator contains a generic relative percentage field for secondary metrics, which is inappropriate for logarithmic SINR values. Its original statistics are retained. A separate reporting pass sets only those SINR percentage fields to null and records the original hash; absolute dB differences, other estimates and every interval are unchanged. This paper uses absolute SINR differences and reports that correction explicitly.

A. Scientific Interpretation

The Study I comparison supports repairing the written incentive in the shared reimplement. Its observation,

dynamics and radio repairs were not independently ablated, so reward cannot be identified as the sole cause of the original thesis's behavior. The Study II result concerns a richer controller: a braking prior, more resource choices, a prospective filter and a wider training range. Its two arms isolate the communication cost in that architecture; their results cannot establish a causal reward effect relative to Study I.

The corrected endpoint excludes every sampled RSS or buffer violation from success. It does not require curved paths, minimum throughput, a packet deadline or an empty arrival queue. Dense stored RSS coverage makes direct trajectories plausible. All successful paths meet sampled constraints, while

substantial delay and unsuccessful missions remain possible. The source reciprocal tradeoff can favor fewer handovers at the expense of delay, as observed in Study II.

Dynamic load, prediction error, switching interruptions, finer temporal resolution, calibrated energy and independent maps remain untested. Prospective map access and the motion prior are control components, not learned capabilities. An application latency or complete delivery requirement would define a new study, and must not be retroactively imposed on these frozen results.

B. Availability

The unified manuscript, source, configurations, raw evaluation records, analysis code, figures, tests, source snapshots and Markdown research notes are provided in the project repository: https://github.com/unworthyzeus/UAV_joint_handover_trajectory_optimization. The private HDF5 and trained checkpoint binaries remain local and are excluded from Git. Reproduction therefore requires authorized access to the map and retraining or separately obtaining checkpoints; this repository is not a self contained public dataset release. Prior separate manuscript versions are retained as historical records; this paper is the consolidated scientific report.

VIII. CONCLUSION

Reward replacement restores destination completion within Study I, whereas arrival termination alone does not. An audit of those positive results exposes the gap between reaching

the destination and maintaining the required connection. Study II repairs the sampled RSS and queue endpoint and restores all three communication objectives. Its full reward lowers weighted cost but increases delay, and its completion advantage over the arrival arm is uncertain. The simpler radio selection reference has the highest observed joint completion among the tested controllers. The result is a documented formulation repair with useful deterministic evidence and mixed learned results, not a complete replication of the original thesis or a general PPO superiority claim.

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APPENDIX A
ORIGINAL TFM AND STUDY I: COMPLETE MODEL AND PPO COMPARISON

The tables compare the written TFM with Study I. Study II retains these settings except for the explicit changes in Appendix B. “Not specified” is an evidence boundary, not an assertion about missing source code. All shared changes apply even to our legacy arm. Only reward replacement and training arrival termination are treatment factors. The exact observation layouts appear in Appendix C. The accompanying note 25 also indexes source definitions and event ordering.

TABLE VII

ENVIRONMENT, OBJECTIVE, AND EVALUATION COMPARISON. SOURCE LOCATORS REFER TO PRINTED TFM PAGES. ROWS DESCRIBE SHARED CHANGES UNLESS MARKED AS FACTORS.

Item	Original TFM definition	Study I implementation
Radio environment	Barcelona, 2.1 GHz, 100 m, 133 sites, nominal 1 m grid; strongest sector RSS [1, pp. 6–7, Sec. 3.1, Table 2]	Received map retained; 87 station operator retained. No ray tracing regeneration; exact source version and sentinel remain unverified.
Lookup and geometry	3D building propagation geometry; lookup and integer encoding not specified [1, pp. 6–7, Sec. 3.1]	Native coordinate spacing, nearest lookup, float radio arithmetic; –128 treated as absent path. No collision model.
Motion	Scalar speed/selected heading update; five acceleration and eight heading bins [1, p. 9, Eq. (2); pp. 12, 15]	Inertial velocity, continuous radial/lateral acceleration, norm bound 5 m/s ² , trapezoidal position update, hard speed cap 25 m/s.
Time and speed	1 ms in prose, 0.1 s in table; 2000 timesteps; 100 m/s penalty threshold [1, pp. 14–16, Tables 3–4]	1 s steps, 200 s deadline. New time discretization and speed constraint.
Arrival (factor)	10 m tolerance, stopping intention, no arrival termination [1, p. 9, Sec. 3.2; p. 12, Sec. 5.2; p. 15, Table 3]	Distance ≤ 10 m and speed ≤ 2 m/s. Training termination varied; every evaluation ends at first safe arrival.
Failure handling	Energy termination; boundary/RSS/speed penalties [1, p. 12, Sec. 5.2, Eq. (14)]	Energy exhaustion, boundary crossing, five consecutive RSS outages terminate. Physical failure precedes arrival; a valid deadline arrival succeeds.
Observations	Position, serving BS, RBG, remaining energy, buffer [1, p. 12, Sec. 5.2]	42 scaled features including goal bearing/distance, velocity, time, consumed energy, prior arrival, phases, serving SINR, candidates, and masks.
Network actions	Station and RBG requests, A3 inequality [1, p. 9, Eq. (3); p. 12, Sec. 5.2]	Stay plus top four RSS stations; > 3 dB advantage and > -96 dBm; five second decisions plus observed outage requests; no time to trigger counters.
Resource allocation	Policy requests RBG; availability fallback; random occupancy of at least half the groups [1, pp. 9, 12, 14–15]	RBG is not learned. First free group, exactly six occupied groups per station in a deterministic pattern; global label phase is not independent load variation.
Rate and interference	Bandwidth plus log of SNR; neighboring RSS sum labeled uplink [1, p. 10, Eqs. (7)–(9)]	Linear downlink cochannel interference, SINR, $B \log_2(1 + \text{SINR})$. Retain 1.44 MHz and -103.41 dBm combined noise; change link interpretation.
Queue and traffic	Poisson formulation, ratio delay, 160 KB, 4800 bit handover overhead; conflicting data packet sizes [1, pp. 9–10, Eqs. (4)–(6); pp. 14–15]	Deterministic 200 kbit/s, decimal 1.28 Mbit cap, same handover overhead. Postservice backlog/rate censored at 200 s; record overflow without terminating.
Energy	Mass/efficiency/lift-to-drag expression; capacity labeled 1000 kW [1, p. 10, Eq. (10); p. 15, Table 3]	Consumed energy integral of $100 + 0.4v^2$ W; 100 kJ budget. New uncalibrated proxy, not a conversion of original capacity.
Reward (factor)	Positive reciprocals, binary +12 progress, penalties [1, p. 12, Eqs. (13)–(14); p. 15, Table 3]	Legacy structure uses new common physical inputs and no soft speed penalty. Replacement adds first arrival/terminal failure payments, time cost, bounded radio costs, and terminal corrected potential shaping.
Deterministic reference	Constant speed, closest discrete heading, viability counters [1, pp. 13–14, Sec. 5.4]	Continuous desired velocity tracking with braking; strongest admissible station and shared masks. A new reference, not a reproduction of that greedy policy.
Splits and comparison	PPO/greedy and weight comparisons; separate validation and test sets not specified [1, pp. 15–16, Secs. 6.2–6.3]	4096 training, 64 validation, 200 standard test, 200 longer test routes; five seeds per treatment, frozen budget. Same map, not a geographic holdout.
Reported outcomes	SNR, outage, interference, remaining energy, handovers, example paths [1, pp. 16–20, Sec. 7]	Mission and recorded feasible success first; secondary metrics conditional on success; every seed, paired bootstrap and failure counts retained.

For completeness, our reset sets velocity, queue, consumed energy, counters, and prior arrival to zero, associates with the strongest station, and allocates its first free group. Each step applies the requested admissible handover before motion, evaluates radio after motion, updates queue and energy, checks failure and arrival, and then computes reward. The TFM does not fully specify this event ordering [1, pp. 9–12, Secs. 3.2–5.2]. The direct energy and packet overflow reward terms removed during the original development [1, pp. 20–21, Sec. 7.3] are not treated as part of its final reward. We also do not claim to have reproduced its mass or efficiency model, random traffic, counter based handovers, or learned resource allocation.

TABLE VIII

PPO COMPARISON. THE FIRST ELEVEN ROWS COVER EVERY PARAMETER IN THE ORIGINAL TABLE 4 [1, p. 16]. FURTHER IMPLEMENTATION DETAILS ARE NOT FULLY SPECIFIED IN ITS SECS. 5.3 AND 6.2 [1, p. 13; pp. 15–16].

Parameter	Original TFM	Study I implementation
Episodes	300	Fixed interactions, variable number of episodes
Episode timesteps	2000	200, with explicit early terminals
Rollout steps per environment	8	128
Learning rate	$3 \cdot 10^{-5}$	$3 \cdot 10^{-4}$, constant
Batch size	32	512
Discount	0.99	0.99, retained
GAE lambda	0.95	0.95, retained
Policy clip	0.2	0.2, retained
Value coefficient	0.5	0.5 applied to half MSE; total coefficient 0.25 MSE
Entropy coefficient	0.01	0.005
Target KL	0.03	No threshold; log diagnostic KL only
Implementation	Stable Baselines	Native PyTorch hybrid PPO
Parallel lanes and update epochs	Not specified	64 lanes; 8192 samples per rollout; four epochs
Interaction budget	Episodes/timesteps listed	524,288 per policy, 64 rollout updates
Networks	Not specified	Separate actor/critic, each two 64 unit tanh layers
Policy heads	Discrete action vector listed	2D Gaussian plus masked five choice categorical
Initialization and variance	Not specified	Orthogonal weights; log standard deviation initialized -0.5 , clamped to $[-2.3, 0.5]$
Optimizer and gradient cap	Adam in pseudocode	Adam, epsilon 10^{-5} , gradient norm cap 0.5
Likelihood and entropy details	Not specified	Latent Gaussian likelihood after stored action sampling; sum latent Gaussian and categorical entropy
Reward and advantage scaling	Not specified	Running discounted return variance; no reward centering; reward clip $[-10, 10]$; minibatch advantage normalization
Critic and bootstrap details	Not fully specified	Uncropped half MSE; bootstrap only across nonterminal rollout ends, not across mission/deadline terminals
Evaluation and checkpoint rule	Not specified	Gaussian mean and categorical mode; final checkpoint only; common raw evaluation reward for every treatment

APPENDIX B

EVERY EXECUTED CHANGE FROM STUDY I TO STUDY II

This inventory compares final frozen implementations, including choices that were retained. Original locators refer to the written TFM, not recovered code. The inherited serialized field `max_consecutive_outage: 5` is unused by Study II: its executed step fails on the first violation. Remaining queue at arrival and instantaneous switching are retained limitations.

TABLE IX: Final Study II differences and original source relationships.

Item	v1 executed behavior	Final v2 executed behavior	Original source relationship
Minimum RSS	$\text{RSS} \leq -96$ dBm counts as outage.	$\text{RSS} < -96$ dBm counts as violation; equality is allowed.	Follow C2's $\geq$ inequality [1, p. 11, Eq. (12)] and the numeric minimum [1, p. 15, Table 3].
Initial RSS	Strongest station selected at reset; no separate persistent initial violation flag.	Check the initial serving RSS and preserve any violation until failure.	Include the initial modeled sample in C2 [1, p. 11].
Failure rule	Five consecutive sampled outages terminate a flight.	Any single sampled RSS violation terminates a failed joint mission.	Enforce the written C2 at modeled samples [1, p. 11], rather than reproduce the source's soft reward penalty [1, p. 12, Eq. (14)].
Buffer overflow	Count lost bits, clip queue, continue flight; disqualify only the subsequent feasible label.	Any overflow ends failure; clipping cannot erase it.	Enforce C4 at the sampled queue update [1, p. 11, Eq. (12)]. Source removal of a direct dump penalty is described on pp. 20–21, Sec. 7.3.
Joint success	Arrival is primary; recorded feasibility permits up to 5% RSS outage and zero drops.	Arrival plus no prior sampled RSS violation, no overflow, and no physical failure. No additional permissive feasibility label.	Combine source communication constraints [1, p. 11] with explicit destination completion.
Event precedence	Some physical failures override arrival; overflow does not.	RSS, overflow, boundary and energy failure override simultaneous arrival.	New explicit ordering; original source ordering is unspecified [1, p. 12, Sec. 5.2].
Outstanding queue at arrival	Allowed.	Still allowed and reported.	The source does not specify an empty queue arrival test [1, pp. 9–12, Secs. 3.2–5.2]. We do not claim all generated data has been delivered.

Continued from previous page

Item	v1 executed behavior	Final v2 executed behavior	Original source relationship
Delay cost	$0.05 * \min(D, 1)$ inside fixed reward.	$0.35 * 10D / (1 + 10D)$ in full arm. No flat reward plateau at one second.	One minus original reciprocal utility [1, p. 12, Eq. (13)] with equal configuration values [1, p. 15, Table 3]. Smooth saturation remains.
Interference cost	Absent from fixed reward, although measured.	Restore $0.30 * 100000I / (1 + 100000I)$.	Same reciprocal utility scales and weights [1, pp. 12, 15]. I remains our downlink proxy.
Handover cost	$0.05 * H$.	$0.35 * 100H / (1 + 100H)$.	Restore normalized source utility [1, p. 12, Eq. (13); p. 15, Table 3].
Positive per step radio utility	Present in reimplemented legacy reward, removed in v1 fixed reward.	Subtract communication cost, so simply staying active does not earn that utility bonus.	Changes the positive reciprocal reward structure [1, p. 12, Eqs. (13)–(14)] while preserving the three normalized tradeoff components.
Learning comparison	Four reward/termination arms; direct acceleration learning.	Two arms: arrival base only versus arrival base minus all three radio costs. Both use strict failure, termination and the same new controller architecture.	A new controlled comparison; neither arm is the source's trained policy.
Network decision timing	Every five seconds, with observed outage exception.	Every modeled second.	Source describes choices each environment step [1, pp. 9, 12, Secs. 3.3.1, 5.2]; a five second interval was our prior assumption.
Network action	Stay plus four strongest candidate stations: five categorical choices.	Stay plus five candidate station slots by 12 groups: 61 choices.	Restores resource choice from the source's BS/RBG action [1, p. 12, Sec. 5.2].
Station candidate selection	Serving plus strongest four, stable RSS ordering.	Retained. Duplicate serving station slots are masked.	Restricted candidate interface is ours; source lists full BS/RBG choices [1, p. 12].
A3 condition	Candidate $RSS > \text{serving RSS} + 3$ dB.	Retained; prospective filter cannot bypass it.	Follow the inequality, not the reversed nearby prose [1, p. 9, Eq. (3); p. 15, Table 3].
Resource allocation	Deterministic first free RBG at reset and handover.	That initialization remains; the controller can choose any currently free RBG, including within the same station.	Source allows RBG requests with fallback [1, p. 9, Sec. 3.3.1]. The exact masking implementation is ours.
Repeating a resource choice	No learned resource choice.	Current serving station/group pair remains a valid pair action; explicit stay also remains.	Avoid a new artificial alternating action mask; this is an implementation decision.
Resource change	Not independently reported.	Count every station/group change, including same station RBG changes.	Diagnostic addition. Same station resource changes are not counted as handovers.
Switching interruption	Not modeled.	Still not modeled; no extra RBG switching cost.	Source does not provide a calibrated interruption duration [1, p. 9, Sec. 3.3.1; p. 15, Table 3]. This remains a limitation, not a verified original behavior.
Continuous action	Two direct radial/lateral acceleration commands.	Projected reference command plus two bounded learned residuals of scale 0.5, followed by physical projection.	New controller prior; differs from original discrete acceleration and heading [1, p. 12, Sec. 5.2; p. 15, Table 3].
Safety filter	None.	Predict the next sample; repair unsafe network choice if possible, otherwise try local motion/network alternatives; still fail if unresolved.	New model based constraint handling component; original source describes reward penalties [1, p. 12, Eq. (14)].
Prospective radio access	Learned policy observes current candidates.	PPO still observes current options. Its shared filter and model based references query the known map at predicted positions.	New information/control architecture, not reproduced source logic.
Observation	42 features.	156 features, detailed below.	Original source has six listed state elements [1, p. 12, Sec. 5.2]. Both are reimplementations with repaired state.
PPO network head	Five categorical logits.	61 categorical logits; continuous and value heads retained.	Original architecture details are not sufficiently specified [1, p. 13, Sec. 5.3].
Training route lengths	200–1000 m.	200–1800 m; longer tests lie within this range.	New study design; original route distribution and independent splits are not documented [1, pp. 15–16, Secs. 6.2–6.3].
Final route identities	Seeds 42012 and 42013.	Fresh seeds 52012 and 52013. Exact pairs are disjoint from v1 and other v2 splits.	Our explicit split protocol; same map remains shared.
Training seeds	1101–1105.	2101–2105.	New repeated experiment; original seed list is unspecified [1, pp. 15–16].
Final budget	524,288 interactions each; 20 policies.	Same interactions per policy; 10 policies.	Our PPO budget remains different from 300 episodes and 2,000 maximum steps in the source [1, p. 16, Table 4].
Deterministic references	Destination braking and strongest admissible RSS.	Four nominal control laws with the same filter: Goal RSS, Goal radio, one step joint search, and three step joint search.	New references, not a recreation of the thesis's discrete constant velocity greedy controller [1, pp. 13–14, Sec. 5.4].
Secondary comparison	Success conditioned summaries and selected matched comparisons.	Explicit intersections of successful route/seed pairs for radio comparisons, with counts and crossed bootstrap intervals.	Source does not report a matching conditional joint success protocol [1, pp. 16–20, Sec. 7].

APPENDIX C

EXACT OBSERVATION AND NUMERICAL INTERFACES

All indices are zero based. Study I has 42 features; Study II has 156. Both replace the original listed position, station, resource, energy and queue vector [1, p. 12, Sec. 5.2]. Scales are fixed; no fitted observation normalizer is used. Study II removes the prior arrival flag, consecutive outage count and five second phase because arrival and RSS failure now terminate immediately.

TABLE X: Study I observation indices and scales.

Indices	Features and scaling
0–1	Position divided by (5000, 3500)
2–3	Unit vector toward goal; (1, 0) fallback at effectively zero distance
4	Goal distance / 1000 m
5–6	Velocity projected onto radial and lateral directions / 25 m/s
7	Speed / 25 m/s
8	Elapsed steps / 200; determines remaining time
9	Queue bits / 1,280,000
10	Consumed energy / 100,000 J, rather than remaining energy
11	Whether first safe arrival already occurred
12–13	Resource label phase / 11 and allocated RBG / 11
14	(step mod 5) / 5
15	Serving SINR in dB / 30
16	Consecutive outage samples / 5
17–21	Five candidate RSS values, each transformed as (RSS + 70) / 60
22–31	Candidate station x/y offsets from UAV, both divided by 5000 m
32–36	Candidate indices within selected operator / 86, not global station IDs or one hot vectors
37–41	Five Boolean admissibility flags represented as float features

TABLE XI: Study II observation indices and scales.

Zero based indices	Definition
0–1	Position / map dimensions
2–3	Destination direction unit vector
4	Destination distance / 1,000 m
5–6	Radial and lateral velocity / 25 m/s
7	Speed / 25 m/s
8	Elapsed decision count / 200
9	Queue / 1,280,000 bits
10	Consumed energy / 100,000 J
11–12	Background phase / 11 and serving RBG / 11
13	Serving SINR in dB / 30
14–18	Candidate RSS, transformed as (RSS + 70) / 60
19–28	Candidate station x/y offsets / 5,000 m
29–33	Candidate station index / 86
34–94	$\log(1 + \text{option_capacity} / 200000)$ for all 61 network choices
95–155	Corresponding Boolean action masks as float features

For station index k , episode label phase z , and group g , background occupancy is $\mathcal{K}[(g - (7k + z)) \bmod 12 < 6]$. The reset allocator chooses $(7k + z + 6) \bmod 12$. Phase z rotates labels globally; it does not yield independent load fields. Each vector lane is an independent single UAV mission. The source describes random occupancy of at least half the groups [1, p. 14, Sec. 6.1; p. 15, Table 3].

Both studies use postservice backlog divided by service rate, censored at 200 s, as delay. A zero service rate is handled numerically rather than interpreted as a measured packet latency. The shared queue generates offered bits and any handover overhead, serves the queue at the modeled capacity, records excess above its capacity and clips for storage. Study II additionally fails on that excess. Stored -128 RSS supplies zero power; SINR reporting has a 10^{-15} logarithm floor. These declarations complement the original queue and rate expressions [1, pp. 9–10, Eqs. (4)–(9)].

Study I evaluates every checkpoint using the common fixed raw reward by default; its stored return is an undiscounted evaluation sum, not its PPO normalized training return. Success and physical metrics define the reported comparisons. In continuing training, first arrival pays its bonus once; a later failure cannot pay the prearrival failure penalty. Study II always terminates on joint success or first failure and scores the proposed residual/network action before environment projection and safety filtering. Neither study has a separate soft speed penalty, since the shared dynamics impose the cap.

APPENDIX D

DEVELOPMENT EVIDENCE AND ARTIFACT MAP

Development results below use only 64 standard and 32 longer validation routes. Each row uses one seed per arm and is exploratory; it does not estimate final seed uncertainty. All six Study II pilot checkpoints and source snapshots are retained locally. Study I retains six pilots with seeds 41 and 42, including the validation curve in Fig. 1. No pilot checkpoint initializes a final policy.

TABLE XII: Study II pilot development; percentages are standard / longer joint success.

Design	Pilot seed and budget per arm	Arrival arm standard / longer joint success	Full arm standard / longer joint success
Direct motion, no filter, short training lengths	51; 524,288	89.1% / 59.4%	92.2% / 34.4%
Direct motion, safety filter, short training lengths	52; 1,048,576	92.2% / 78.1%	98.4% / 75.0%
Residual motion, safety filter, 200–1800 m training lengths	53; 524,288	96.9% / 84.4%	98.4% / 87.5%

The first strict nominal RSS controller completed 95.3% / 71.9% of validation routes; prospective radio selection completed 96.9% / 90.6%. A local joint search initially completed 95.3% / 81.3% and sometimes preferred stationary low cost states. A preference for immediate progress, with the shared filter, raised its observed validation completion to 98.4% / 87.5%. The filtered radio reference reached 96.9% / 93.8%. A three step lookahead design initially postponed movement at each replan, reaching 70.3% / 53.1%; applying progress preference to the first predicted step yielded 93.8% / 87.5%, still with three standard timeouts. These development failures are retained, not hidden by the final controller selection.

In the second pilot design, the current station/group pair remained legal in addition to explicit stay, eliminating an artificial alternating resource mask. The third design introduced the residual braking prior and 200–1800 m training range. No communication threshold was relaxed. The full final pilot already showed fewer handovers and higher delay, motivating explicit component reporting in the frozen comparison. These simultaneous development changes do not identify individual causal contributions.

The following artifact families give a reproducible route from source data to this paper. They are paths relative to the repository root; private binaries are retained locally. Markdown notes 17–27 cover Study I and its audit, notes 28–32 cover Study II, and note 33 records this consolidation and publication.

TABLE XIII: Artifacts and reproducibility boundaries.

Artifact	Location / purpose
Original map	dataset/Barcelona_dataset_January.h5; unchanged private HDF5, excluded from Git; SHA256 is stored in both freeze manifests.
Study I freeze	configs/frozen_comparison_v1.json; six exact source/protocol hashes.
Study II freeze	configs/frozen_connectivity_v2.json; seven exact source/protocol hashes.
Routes	configs/controlled_scenarios.json and configs/connectivity_scenarios_v2.json; saved coordinate pairs, phases and seeds.
Outcomes	results/controlled_experiment/confirmatory_v1/ and results/connectivity_experiment/confirmatory_v2/; final evaluation records, training logs and source snapshots.
Statistics	Corresponding analysis_v1/ and analysis_v2/ directories; raw estimators, paired intervals, generated figures and table inputs. Study II report_statistics.json documents the SINR reporting correction.
Replay	results/controlled_experiment/connectivity_audit_v1/ and results/connectivity_experiment/replay_v2/; 4,400 audit and 5,600 final exact replays respectively.
Paper	paper/unified.tex and paper/unified_appendices.tex; shared generated tables and figures; final consolidated IEEE format draft.